UNIT-1

1.a) Show that (P→(Q→R)) -> ((P->Q) -> (P->R)) is a tautology.

b) Determine whether (~q^ (p->q))-> ~p is a tautology?

2. a) Compare and contrast the different types of connectives used in mathematical logic.

b) Prove that ∀x(P(x) →Q(x)), ∀x(R(x) →~ Q(x)) ⇒ ∀x(R(x) -> ~ P(x)).

3.a) "All lions are fierce", "Some lions do not drink coffee", "Some fierce creatures do not drink coffee". Express the statements using quantifiers p(x)

b) Translate each of the following statement into symbols, using quantifiers, variables and predicate symbols.

(i) All birds can fly

(ii) Some babies are illogical

(iii) Some men's are giants

(iv) There is a student who like mathematics but not history.

4.a) Construct the truth table for (p→ q) ↔ (~p V q).

b) Construct the truth table for [(P^Q) V (~P^R)] V (Q^R).

5 a) a) Prove that √2 is irrational.

(b) Show that ~p is a valid conclusion from premises.

p-> r, r-> s, t v ~s, ~t v u, ~u.

6.a) Obtain the principle disjunctive normal form (P^Q) V (~P v R) V (Q^R).

b) Show that the given set of premises are inconsistent

p→ q, q→ r, s→ ~r, p^s

Unit-II

1. Explain about Composite relation with an example.
2. Consider the following relations on {1,2,3,4)

R1 = ((1,2), (1,1), (2,1), (2,2), (3,4), (4,1), (4,4))

R2 = {(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4.4))

R3 = ((2,1), (3,1), (3,2), (4,1), (4,2), (4,3)}

R4= {(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4))

Which of these relations are equivalence relations?

3. Let A={1,2,3), B={1,2,3,4) the relations R1 = {(1,1), (2,2), (3,3)) and R2 = {(1,1),(1,2), (1,3), (1,4)} and obtain (a) RI U R2 (b) R1 ^ R2 (c) R1-R2 (d)R2-R1
4. What is the Cartesian product A X B X C, where A ={0,1} B={1,2} C = {0, 1, 2} ?
5. Let A = {1,2,3,4}and and let R={(1,1), (1,2), (2,1),(2,2), (3,4), (4,3), (3,3), (4,4)} be partial order relation on R ? determine A/R.
6. a) Define transitive closure of a binary relation.

   b) By using Warshall's algorithm, find the transitive closure of the relation R={(2,1),(2,3),(3,1),(3,4),(4,1),(4,3)} on set A={1,2,3,4}.
7. Obtain the zero-one matrix of the transitive closure of the relation R where

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

8. Explain the closure of relations.
9. Explain types of reations with example.